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SentinelEye: ML-Based Wild Animal Intrusion Detection System Using Raspberry Pi.

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Abstract:

The present invention relates to a real-time wild animal intrusion detection and monitoring system designed for deployment in agricultural lands, forest-border areas, public spaces, institutional campuses, residential zones, highways, and other regions vulnerable to wild animal intrusion. The proposed system is divided into two integrated modules: an edge-based intrusion detection prototype using Raspberry Pi and a desktop-based web monitoring system developed using Flask.

The edge-based prototype utilizes a Raspberry Pi integrated with a Pi Camera module for continuously capturing live video streams. Captured frames are processed in real time using OpenCV and a pretrained deep learning model based on the **ResNet18** architecture for animal detection and classification. Upon detecting a wild animal such as a leopard, the system automatically generates localized audio alerts through a Loud speaker to warn nearby individuals and simultaneously sends SMS notifications containing the detected animal name, location, date, and timestamp to registered users and authorities through the **SIM900A** GSM module.

For real-world implementation and centralized monitoring, a Flask-based desktop GUI system is developed consisting of Admin and User panels. The web interface provides functionalities such as live camera feed monitoring, detection log management, user registration, location management, authentication, and user access control. Detection records including animal name, location, date, and time are maintained for monitoring and analysis purposes.

By integrating Machine Learning, Deep Learning, Artificial Intelligence, Computer Vision, IoT communication, and web-based monitoring technologies, the proposed system provides an automated, real-time surveillance and public alert mechanism aimed at reducing human–wildlife conflict and improving public safety.

Keywords: *Wild Animal Intrusion Detection, ResNet18, Raspberry Pi, Flask, Deep Learning, Computer Vision, IoT, Smart Agriculture, Edge AI, GSM Alert System, Real-Time Monitoring, Public Safety.*

I. INTRODUCTION

Wildlife intrusion into human settlements has become a significant global concern, particularly in agricultural regions and areas adjacent to forest boundaries [2], [3], [5]. Farmers frequently incur substantial crop damage and economic losses due to the entry of wild animals such as leopards, wild boars, elephants, and tigers into farmlands [4], [5]. In addition, individuals residing near forest peripheries—including highways, institutional campuses, and residential zones—face increasing safety risks from unexpected animal encounters [7], [8]. Conventional preventive measures, such as fencing, manual surveillance, and watchtowers, are often expensive, labor-intensive, and unreliable under varying environmental conditions, thereby limiting their effectiveness and scalability [2], [5], [11].

Recent advancements in Artificial Intelligence (AI), Computer Vision, and Internet of Things (IoT) technologies have facilitated the development of automated and intelligent surveillance systems [3], [7], [8]. Camera-based monitoring combined with deep learning models enables accurate detection and classification of animal activity in real time [1], [2], [4]. Deep learning architectures such as Convolutional Neural Networks (CNNs) and YOLO-based object detection models have demonstrated significant improvements in image classification and object recognition tasks under diverse environmental conditions [1], [3], [9]. Furthermore, the adoption of edge computing platforms, such as Raspberry Pi, allows local data processing, reducing latency and enabling continuous system operation even in remote areas [4], [10], [11].

In our previously published review study, **“SentinelEye: ML-Based Wild Animal Intrusion Detection System Using Raspberry Pi – A Review”** [11], a comprehensive analysis of various deep learning approaches, including YOLO-based object detection models and CNN architectures, was conducted for wildlife intrusion detection. The review highlighted the advantages and limitations of existing approaches with respect to detection accuracy, computational complexity, response time, and deployment feasibility on low-power embedded devices. Based on the insights derived from this review and considering real-world deployment constraints such as computational efficiency, power consumption, and hardware limitations, the present work adopts a lightweight deep learning model based on the ResNet18 architecture for efficient real-time classification on edge devices [9].

The proposed system, SentinelEye, is designed as a two-module architecture comprising an edge-based intrusion detection unit and a centralized web-based monitoring system. The edge module utilizes a Raspberry Pi integrated with a Pi Camera to continuously capture video frames, which are processed in real time using OpenCV and a pretrained ResNet18 model [1], [4]. The captured frames are analyzed to identify the presence of wild animals, and upon successful detection, the system generates immediate alerts through a Bluetooth speaker while simultaneously sending SMS notifications containing the detected animal type, location, date, and timestamp via a GSM communication module [4], [11]. This real-time alert mechanism assists farmers, forest authorities, and nearby residents in taking immediate preventive actions to minimize potential threats and damages.

In addition to the edge-based detection unit, a Flask-based desktop web application is developed to enable centralized monitoring and system management [8]. The graphical user interface (GUI) consists of separate Admin and User panels, providing functionalities such as live camera feed visualization, detection log management, user registration, location configuration, system monitoring, and role-based access control. The centralized platform improves usability, enhances monitoring efficiency, and maintains historical intrusion records for future analysis and decision-making.

By integrating Edge AI, Deep Learning, IoT communication, and web-based monitoring, the proposed SentinelEye system provides a comprehensive, scalable, and cost-effective solution for real-time wildlife intrusion detection. The system aims to minimize human–wildlife conflicts, enhance agricultural protection, reduce economic losses, and improve public safety across diverse deployment environments. Furthermore, the proposed architecture offers practical applicability for smart agriculture, forest-border surveillance, institutional campus security, and remote area monitoring systems.

II. LITERATURE REVIEW

Several researchers have explored Artificial Intelligence (AI), Computer Vision, and Internet of Things (IoT)-based approaches for automated wild animal intrusion detection. Saxena et al. [1] proposed an AI-based farm security system using OpenCV and deep learning for real-time detection and alerts, though it faces scalability challenges. Mamat et al. [2] implemented a YOLOv5-based model achieving high accuracy (~94%), but with limitations in low-light and multi-camera scenarios. Similarly, Delwar et al. [3] introduced a YOLOv8-based IoT system with very high accuracy, though it requires stable internet connectivity and higher computational resources.

Abraham et al. [4] developed a YOLOv3-based system integrated with Raspberry Pi and GSM modules for real-time alerts, while Panda et al. [5] proposed a sensor-assisted IoT system, which is cost-effective but less accurate compared to vision-based approaches. Patil et al. [6] focused on distinguishing harmful and non-harmful animals, improving reliability but remaining sensitive to environmental conditions. Nandan et al. [7] and Kumar et al. [8] explored AI-based wildlife monitoring and multi-functional systems, though these are more complex and not optimized for real-time intrusion detection.

Recent work by Kant and Yamsani [9] demonstrated high classification accuracy using ResNet152V2, but with increased computational complexity unsuitable for edge devices. Tandale et al. [10] proposed a Raspberry Pi-based portable detection system, which lacks centralized monitoring capabilities. Our previous review study [11] highlighted key challenges in existing systems, including high computational cost, environmental sensitivity, and limited scalability.

Overall, while existing approaches show promising results, limitations in computational efficiency, real-time edge deployment, and centralized monitoring persist. These gaps motivate the proposed lightweight and scalable ResNet18-based intrusion detection system with integrated web-based monitoring.

III. PROBLEM DEFINITION

In recent years, wild animal intrusion into agricultural fields, forest-border villages, highways, and public spaces has emerged as a significant challenge, leading to substantial crop damage, economic losses, and threats to human safety. Animals such as leopards, wild boars, elephants, and monkeys frequently enter human-inhabited areas, resulting in increased human–wildlife conflicts. Existing preventive measures, including electric fencing, manual patrolling, and conventional CCTV-based surveillance, are often expensive, labor-intensive, and ineffective under real-world conditions such as low lighting, occlusion, and large-area coverage.

Although recent advancements in Artificial Intelligence (AI), Computer Vision, and IoT-based systems have improved automated intrusion detection, many existing solutions suffer from key limitations. These include high computational complexity, dependency on stable internet connectivity, reduced performance on resource-constrained edge devices, and lack of integrated real-time alerting and centralized monitoring capabilities.

Therefore, there is a critical need for a lightweight, efficient, and scalable intrusion detection system capable of operating in real time on low-power edge devices. The system should provide accurate animal classification, rapid alert generation, and centralized monitoring support to ensure timely response and improved safety. Addressing these challenges forms the core objective of the proposed system, which leverages a ResNet18-based deep learning model deployed on Raspberry Pi, integrated with IoT-based alert mechanisms and a web-based monitoring interface.

IV. PROPOSED SYSTEM

The proposed system, SentinelEye, is designed as a hybrid architecture comprising two interconnected modules: (i) an edge-based intrusion detection unit deployed on Raspberry Pi and (ii) a centralized web-based monitoring system developed using the Flask framework. The overall objective is to enable real-time wild animal detection, rapid alert generation, and centralized monitoring with minimal computational overhead.

A. Edge-Based Intrusion Detection Module:

The edge module is implemented using a Raspberry Pi integrated with a Pi Camera for continuous video capture. The captured frames are processed using OpenCV, including preprocessing steps such as resizing and normalization.

A pretrained ResNet18 model is used for real-time animal classification. The model is optimized for deployment on low-power devices, ensuring efficient and low-latency performance. Each frame is analyzed to detect the presence of wild animals such as leopards and wild boars.

Upon detection, the system generates instant alerts. A GSM module (SIM900A) sends SMS notifications containing the detected animal type, location, date, and time, while a loudspeaker produces real-time audio warnings for nearby individuals. Additionally, all detection events are stored locally along with image snapshots for monitoring and analysis.

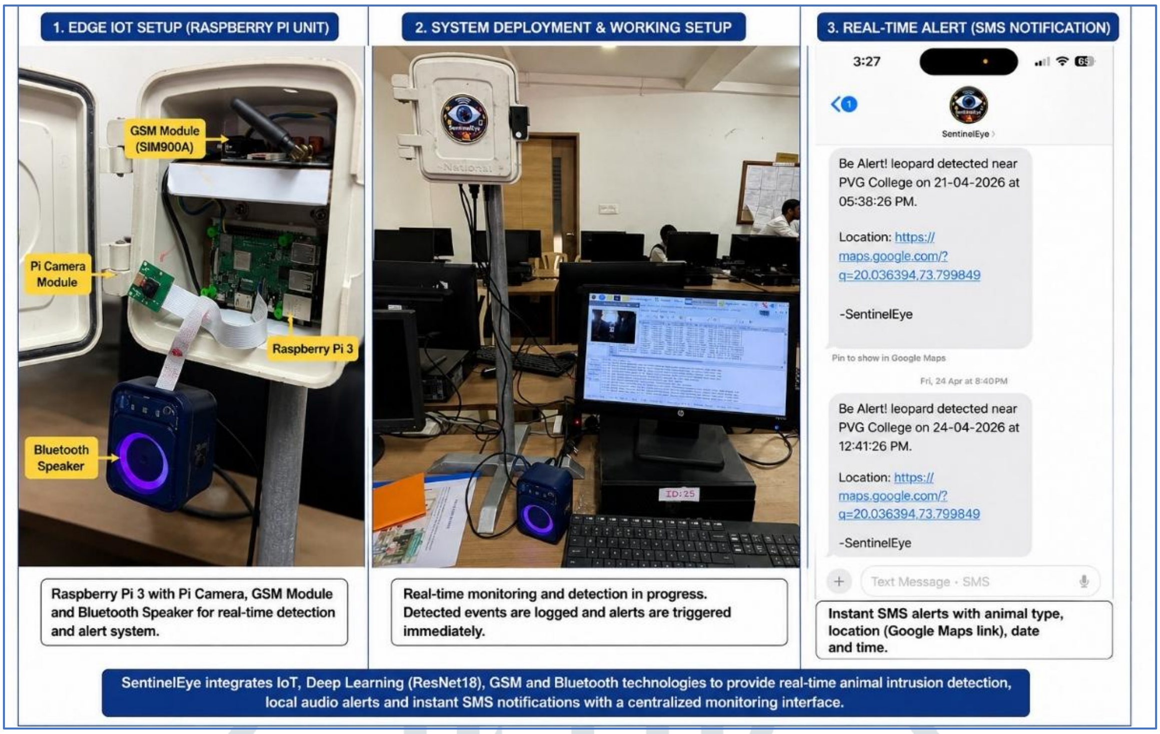

Fig. 1. Edge-Based IoT Hardware Setup for Animal Intrusion Detection

B. Centralized Web-Based Monitoring Module:

A Flask-based web application is developed to provide centralized monitoring and control of the intrusion detection system. The system supports role-based access through Admin and User interfaces.

The Admin dashboard enables real-time monitoring of animal detections, displaying system statistics, live alerts, and camera feed, as shown in Fig. 2. It also maintains structured detection logs containing animal type, location, date, and time (Fig. 3). Additionally, the system settings interface allows configuration of alert mechanisms such as SMS notifications and sound alerts, along with location management (Fig. 4).

The User interface provides registered users with access to live camera feed, alert notifications, and recent detection logs, as illustrated in Fig. 5. This ensures that users can monitor intrusion events in real time and take appropriate action.

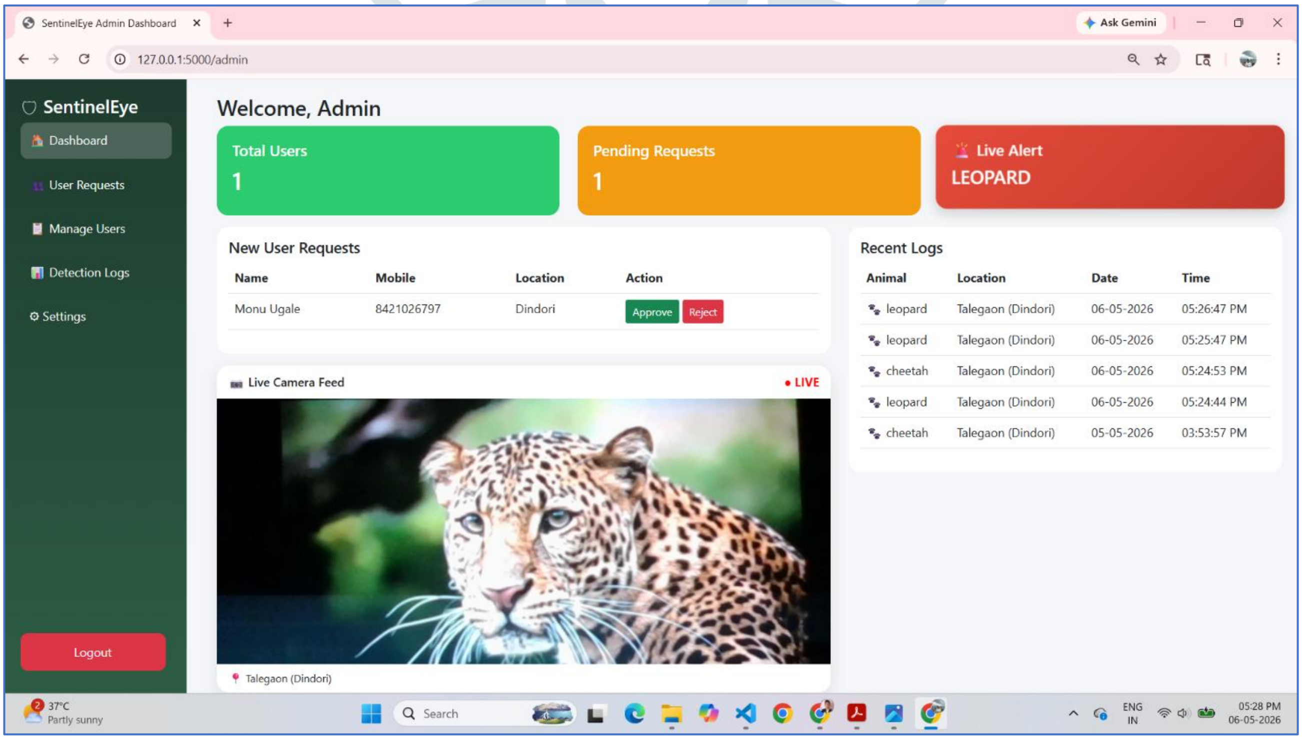

Fig. 2. Admin Dashboard with Live Alerts and Camera Feed

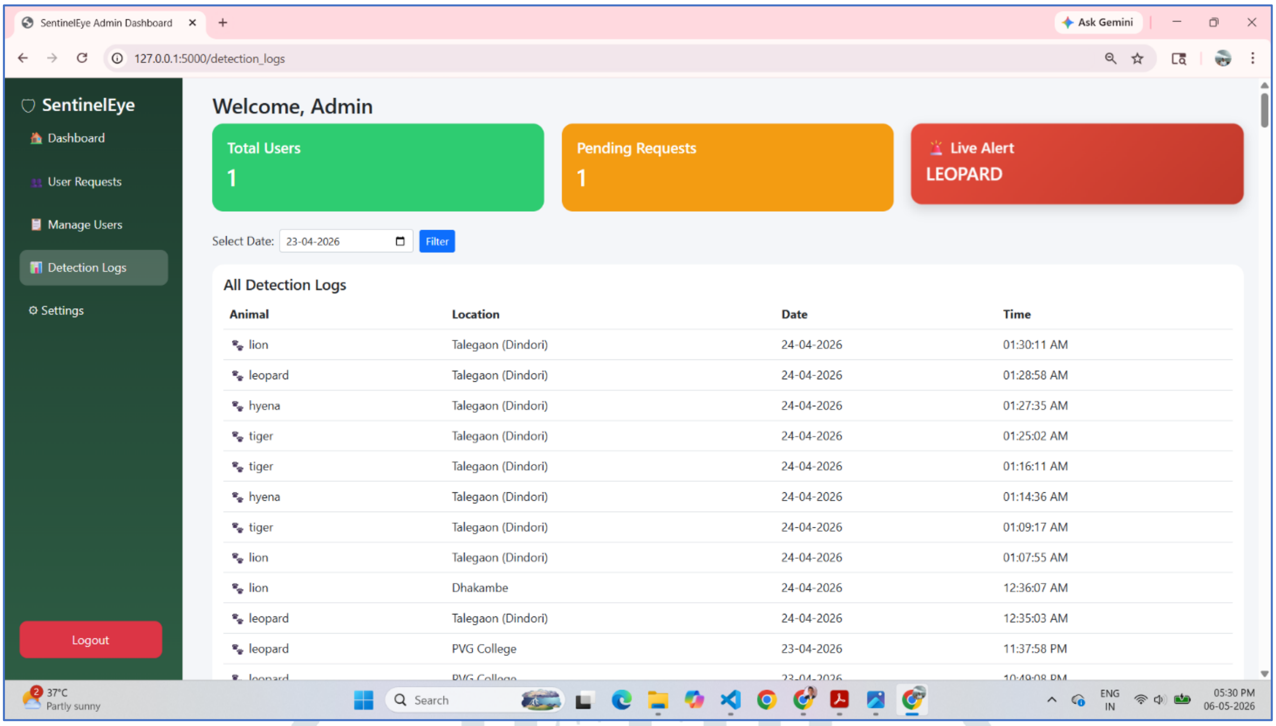

Fig. 3. Admin Login - Animal Detection Logs

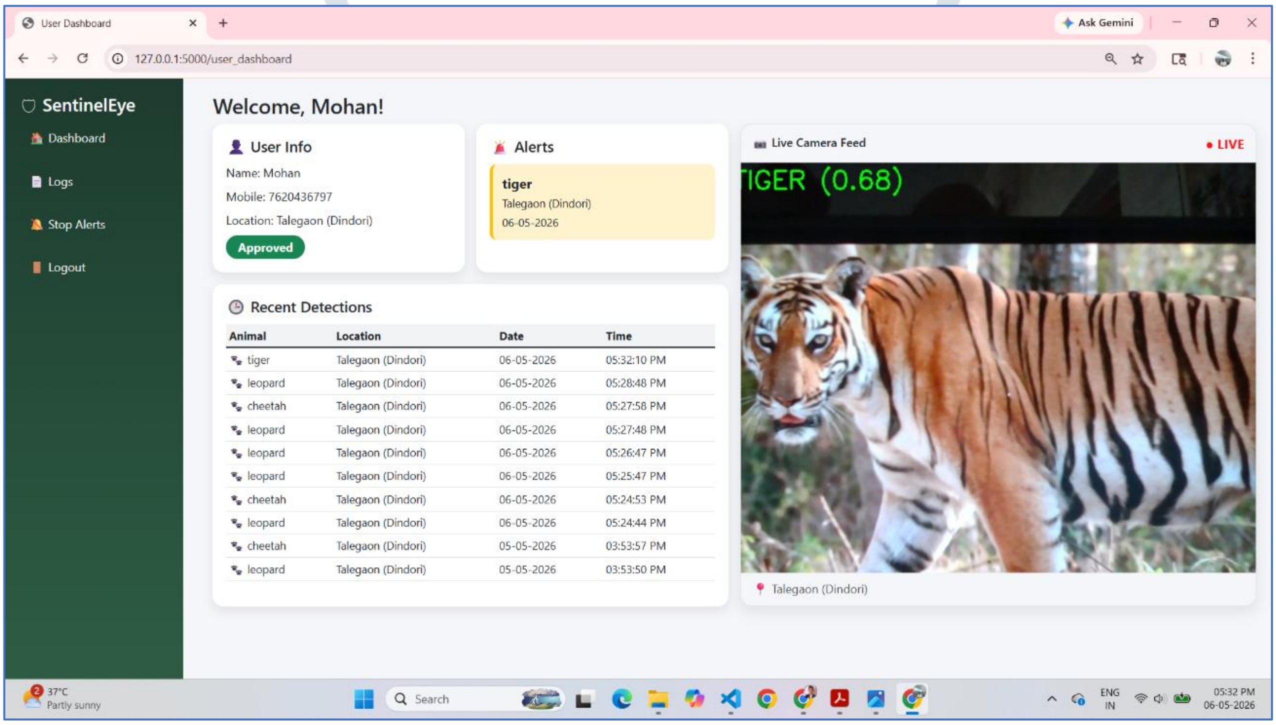

Fig. 4. User Login - User Dashboard with Live Feed and Alerts

C. System Workflow:

The overall workflow of the system is as follows:

1.Continuous image capture using Pi Camera

2.Frame preprocessing using OpenCV

3.Animal classification using ResNet18

4.Intrusion detection decision

5.Triggering alerts via GSM and Bluetooth modules

6.Logging detection data

7.Displaying results on the Flask-based GUI

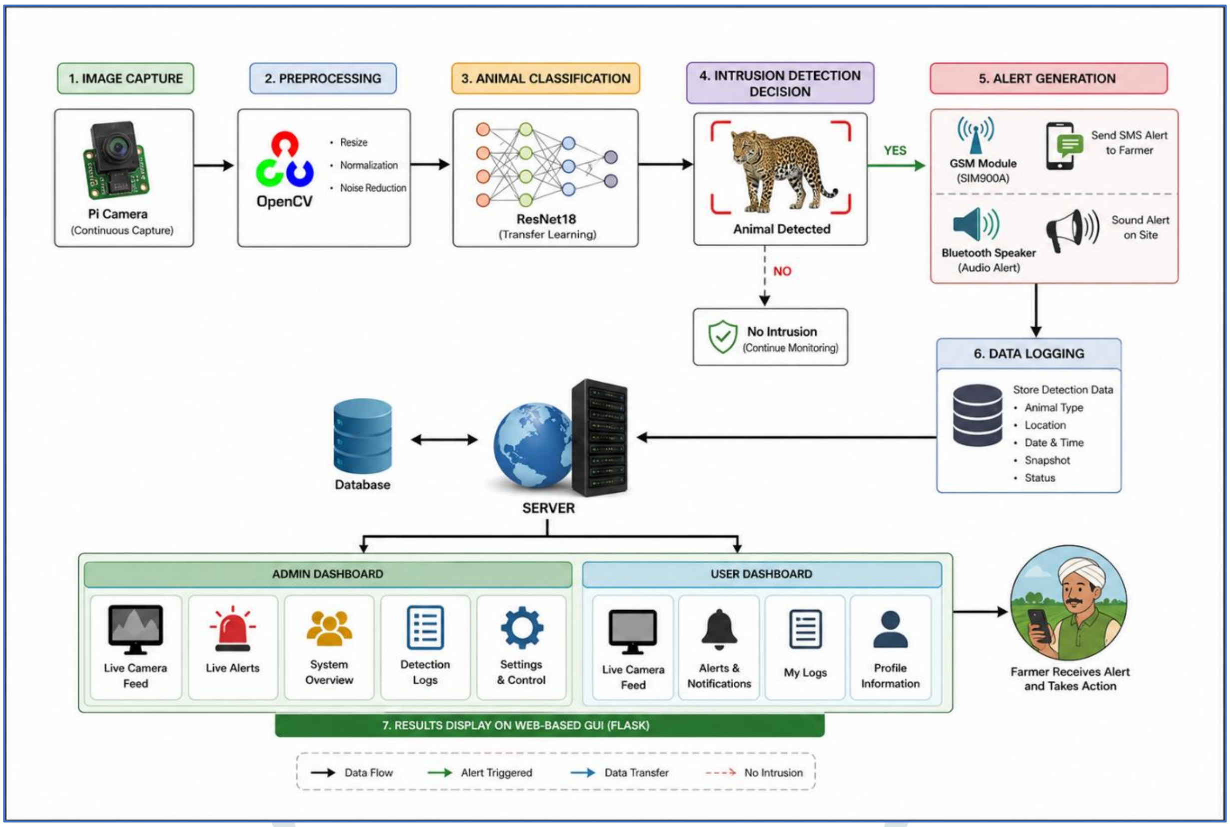

Fig. 5. System Workflow of the Proposed Wildlife Intrusion Detection System

The overall workflow of the proposed system integrates edge-based processing with centralized monitoring to enable real-time wildlife intrusion detection. The system continuously captures images using a Pi Camera and processes them using OpenCV for preprocessing. A pretrained ResNet18 model performs efficient animal classification, and based on the prediction, an intrusion decision is made. Upon detection, alert mechanisms such as GSM-based SMS notifications and audio warnings are triggered. The detected events are logged for analysis and are simultaneously displayed on a Flask-based web interface, allowing both administrators and users to monitor intrusion activities in real time.

D. Model Description:

The proposed system utilizes a pretrained ResNet18 model for real-time animal classification. ResNet18 is a convolutional neural network (CNN) architecture based on residual learning, which enables efficient training of deep networks by addressing the vanishing gradient problem through shortcut connections.

The input image captured from the Pi Camera is first resized to (224 × 224) pixels and normalized before being passed through the network. The convolutional layers extract spatial features from the input image using learned filters. The convolution operation is mathematically defined as:

$$y(i,j) = \sum_{m} \sum_{n} x(i+m, j+n) \cdot w(m,n)$$

To introduce non-linearity and improve feature learning, the Rectified Linear Unit (ReLU) activation function is applied after each convolutional layer:

$$f(x) = \max(0, x)$$

ResNet18 employs residual connections that allow the network to learn identity mappings, which can be expressed as:

$$y = F(x) + x$$

where (F(x)) represents the residual mapping learned by the network. This improves training efficiency and model performance on deeper architectures.

The final fully connected layer produces output scores for different classes, which are converted into probabilities using the Softmax function:

$$P(y=i)=\frac{e^{z_i}}{\sum_j e^{z_j}}$$

The Softmax function enables classification by assigning probabilities to each class. A threshold-based decision rule is applied to determine intrusion detection in real time.

V. EXPERIMENTAL RESULTS

The proposed SentinelEye system was evaluated using real-time video streams captured through a Pi Camera and a test dataset comprising animal classes such as leopard, tiger, lion, hyena, and wild boar. The system was deployed on a Raspberry Pi to assess real-time performance under practical conditions.

The ResNet18-based model achieved a classification accuracy of approximately 92–95% under normal lighting conditions. The average detection time per frame was around 1–2 seconds, while alert notifications were generated within 3–5 seconds after detection. The system successfully triggered both SMS alerts and audio warnings, with results displayed on the Flask-based web interface.

Performance evaluation using a confusion matrix (Fig. 6) shows that most predictions lie along the diagonal, indicating high classification accuracy with minor misclassifications between similar classes. Additionally, the accuracy graph (Fig. 7) demonstrates stable and consistent performance across multiple iterations without significant fluctuations.

These results confirm that the proposed system is capable of reliable, real-time animal intrusion detection with low latency, making it suitable for practical deployment.

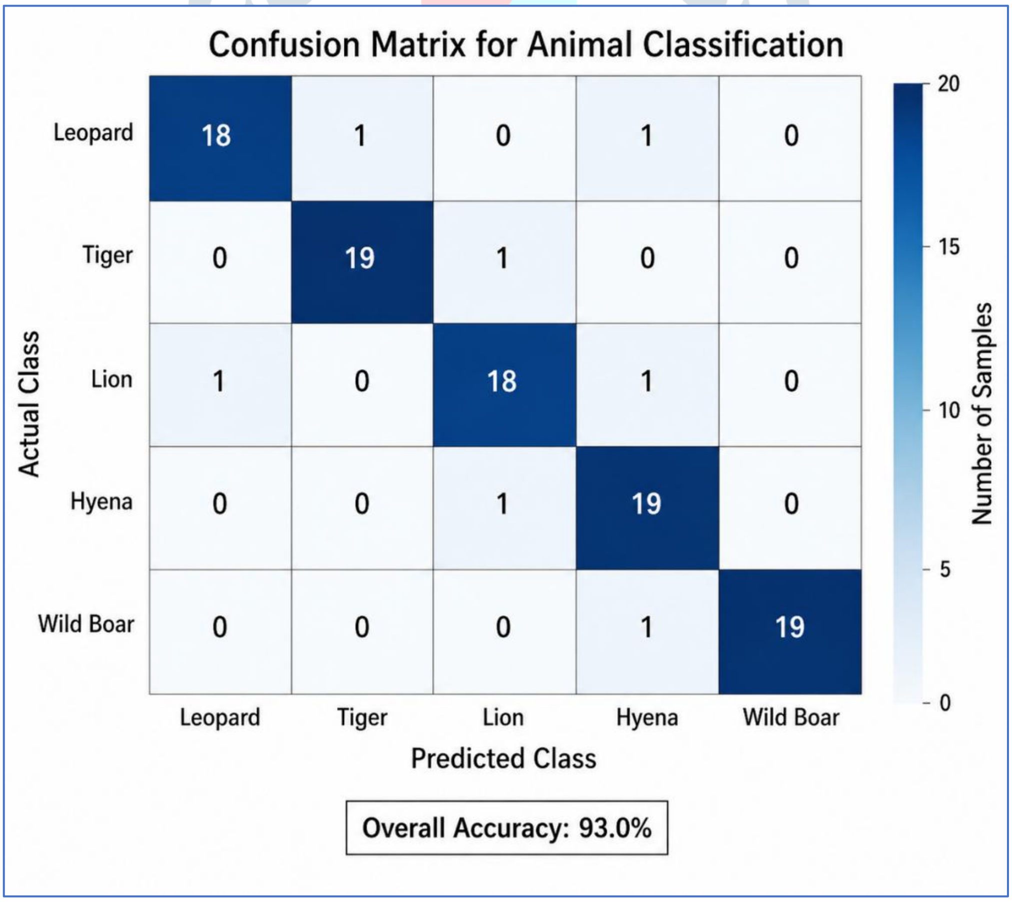

Fig. 6. Confusion Matrix for Animal Classification

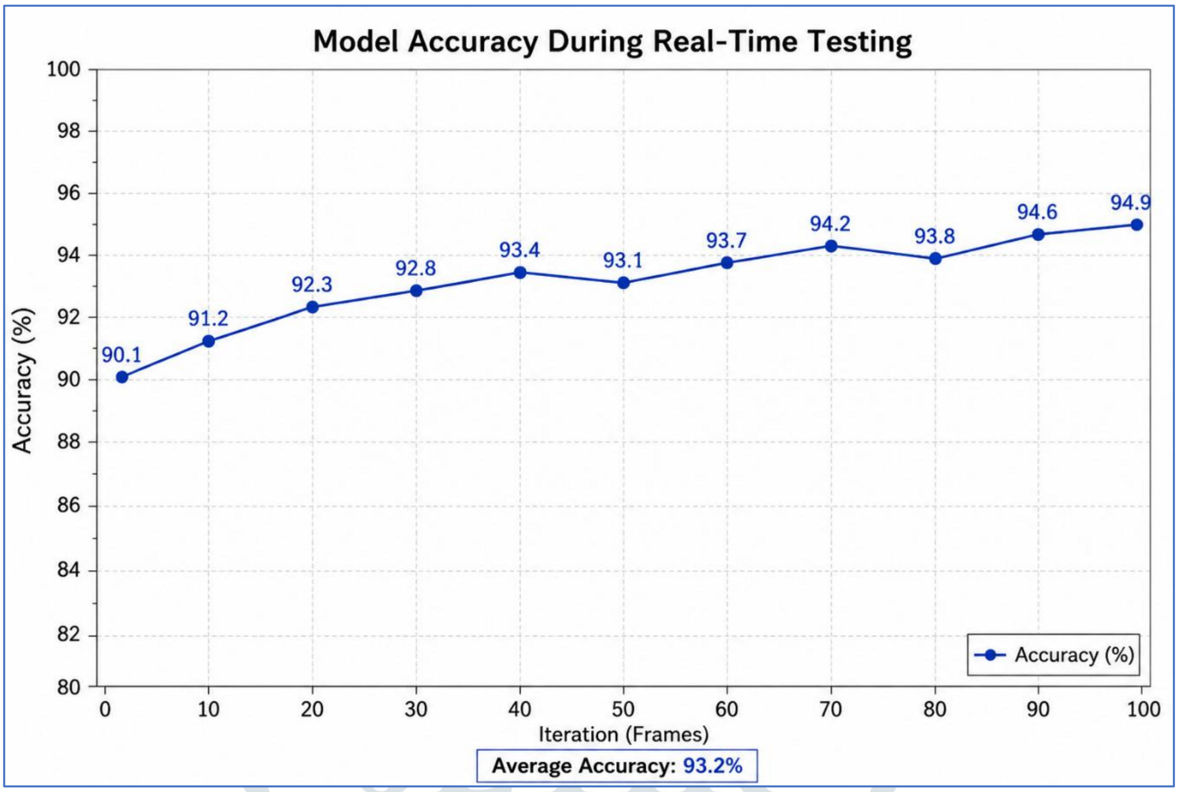

Fig. 7. Accuracy Variation During Real-Time Testing

CONCLUSION

This research presents SentinelEye, a real-time wild animal intrusion detection system that integrates deep learning, computer vision, and IoT technologies to mitigate human–wildlife conflict. The system utilizes a pretrained ResNet18 convolutional neural network for accurate animal classification, deployed on a Raspberry Pi for efficient edge-based processing.

The proposed system captures live video using a Pi Camera, preprocesses frames using OpenCV, and performs real-time classification with optimized inference suitable for low-power devices. Upon detecting target animals such as leopard, tiger, or wild boar, the system generates immediate alerts through SMS notifications and audio warnings. Detection events are stored in a database and visualized via a Flask-based web interface, enabling centralized monitoring and control.

Experimental results demonstrate high classification accuracy and reliable real-time performance with low latency. The integration of edge computing with a web-based monitoring system ensures scalability, cost-effectiveness, and suitability for deployment in rural and semi-urban environments.

The implementation of SentinelEye also highlights the practical feasibility of using affordable embedded hardware for intelligent surveillance applications. By performing classification directly on the Raspberry Pi, the system reduces dependency on high-end servers and continuous internet connectivity. This makes the solution suitable for remote agricultural fields, forest-border villages, highways, and residential areas where conventional surveillance systems are costly or difficult to maintain.

The system also supports effective decision-making by maintaining detection logs, alert records, and monitoring data through the web dashboard. These stored records can help farmers, residents, and authorities analyze intrusion patterns, identify frequently affected locations, and plan preventive actions. The non-lethal alert mechanism promotes safer coexistence between humans and wildlife by deterring animals without causing harm.

Overall, the proposed system provides an efficient, automated, and affordable solution for wildlife intrusion detection, enhancing farm security, reducing economic losses, and improving human safety.

REFERENCES

[1] A. Saxena, A. Shisodia, and D. Upadhyay, “Enhancing Farm Security System with AI-Powered Animal Intrusion Detection Mechanism,” IEEE, 2025.

[2] N. Mamat, M. F. Othman, and F. Yakub, “Animal Intrusion Detection in Farming Area Using YOLOv5 Approach,” IEEE, 2022.

[3] T. S. Delwar, S. Mukhopadhyay, A. Kumar, M. Singh, Y.-w. Lee, J.-Y. Ryu, and A. S. M. S. Hosen, “Real-Time Farm Surveillance Using IoT and YOLOv8 for Animal Intrusion Detection,” Future Internet, 2025

[4] A. Abraham, B. Mathew, D. Panikkar, and J. John, “Wild Animal Intrusion Detection System Using YOLO,” IJISRT, May 2023.

[5] P. K. Panda, C. S. Kumar, B. S. Vivek, M. Balachandra, and S. K. Dargar, “Implementation of a Wild Animal Intrusion Detection Model Based on Internet of Things,” IEEE, 2022.

[6] P. A. Waghmare, S.U. Khabade, P. P. Kokate, Sagar Patil and Suchita Patil “Enhanced Animal Intrusion and Detection System Using Artificial Intelligence,” IJRAR, May 2023.

[7] A. Nandan, S. N. Tiwari, and A. L. Yadav, “AI-Driven Wildlife Monitoring and Protection System”, ICICC, 2024.

[8] K. Likhitha, B. R. Chowdary, C. Praneeth, K. S. Satya, and K. Jairam “IoT-Based Real-Time Wild Animal Detection and Alert System Using Deep Learning and Multi-Sensor Integration,” IRJAEH, 2026.

[9] V. Kant and N. Yamsani, “Animal Species Classification Using Fine-Tuned ResNet152V2: A Deep Learning Approach for Biodiversity Conservation”, IEEE, 2025

[10] A. Tandale, K. Barsagade, N. Wyawahare, P. Saraf, N. C. Morris, and H. S. Zambani, “Smart Stick to Detect Wildlife Animals for Trekkers Using Raspberry Pi”, IEEE, 2025.

[11] M. S. Ugale, S. Y. Patil, A. M. Awalgave, S. S. Sonawane, and C. H. Patil, “SentinelEye: ML-Based Wild Animal Intrusion Detection System Using Raspberry Pi – A Review”, IJARSCT, 2025